

- 6 (a) A cell of e.m.f. 1.5V and internal resistance $0.25\ \Omega$ is connected in series with a resistor R, as shown in Fig. 6.1.

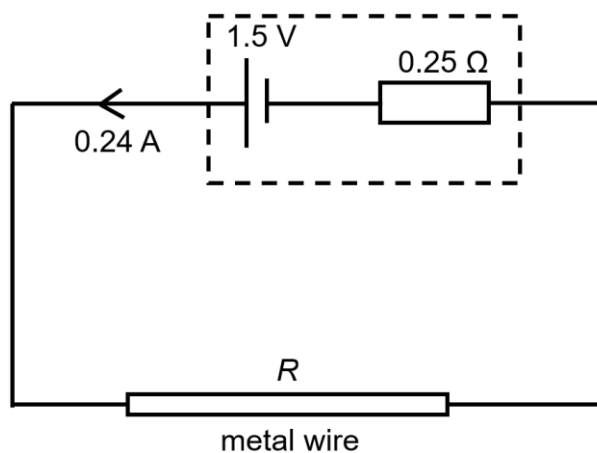


Fig. 6.1

The resistor R is made of metal wire.

A current of 0.24 A passes through R for a time of 5.0 minutes.

Calculate

- (i) the charge that passes through the cell,

$$Q = I t = 0.24 \times 5.0 \times 60 = 72\ \text{C}$$

charge = C [1]

- (ii) the total energy transferred by the cell,

$$W = Q V = 72 (1.5) = 108\ \text{J}$$

Or

$$W = I E t = (0.24)(1.5) (5 \times 60) = 108\ \text{J}$$

energy = J [2]

- (iii) the energy transferred in the resistor R,

By conservation of energy,

Energy transferred in R = Total energy – Energy dissipated in $0.25\ \Omega$

$$= 108 - I^2 r t$$

$$= 108 - (0.24)^2 (0.25)(5.0 \times 60)$$

$$= 104\ \text{J}$$

[2]

(iv) the resistance of R.

$$E = I^2 R t$$

$$104 = (0.24)^2 (R)(5.0 \times 60)$$

$$R = 6.0 \, \Omega$$

$$E = I (R + r)$$

$$1.5 = 0.24 (R + 0.25)$$

$$R = 6.0 \, \Omega$$

resistance = Ω [2]

(b) Two cells identical to the one in (a) are now connected in series with a fixed resistor of resistance $2000 \, \Omega$ and a thermistor, as shown in Fig. 6.2.

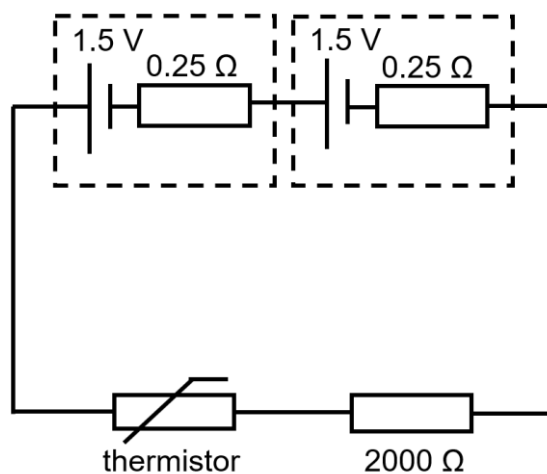


Fig. 6.2

The thermistor has resistance $4000 \, \Omega$ at 0°C and $1800 \, \Omega$ at 20°C .

(i) Explain why, in this circuit, the internal resistance of the cells may be considered to be negligible.

The combined resistance of the fixed resistor and thermistor at 20°C is between $3800 \, \Omega$ and $6000 \, \Omega$,

while the combined resistance of the internal resistance of the cells is $0.50 \, \Omega \times 2$.

Since the external resistance ($5800 \, \Omega$) is much greater than the internal resistance of the cells ($0.50 \, \Omega$), the internal resistance of the cells can be neglected, whereas the external resistance in fig 7.1 is $6.0 \, \Omega$.

[1]

- (ii) In one particular application of the circuit of Fig. 6.2, it is desired that the potential difference across the **fixed** resistor should range from 1.2 V at 0 °C to 2.4 V at 20 °C.

Determine whether it is possible to achieve this range of potential differences.

In order to get that range R would need to be changed to another resistor.

Let the resistor of the fixed resistor be R.

At 0°C, R needs to be 1.2 V

$$\frac{V_R}{E} = \frac{R}{R + 4000}$$

$$\frac{1.2}{3.0} = \frac{R}{R + 4000}$$

$$R = 2700 \, \Omega$$

Checking whether 2700 Ω will give 2.4 V at 20°C,

$$\frac{V_R}{E} = \frac{R}{R + 4000}$$

$$\frac{V_R}{3.0} = \frac{2700}{2700 + 1800}$$

$$V_R = 1.8 \, \text{V}$$

Since the same fixed resistor does not give the required range, it is not possible to achieve this range of potential differences.

[3]

11]

